

THE RAY-CLASS REPAIR FOR AFFINE SYSTEMS: THE CLASS GROUP AS THE INDEX OF DEFINABLE SCALINGS

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ABSTRACT. Paper XXX computed the K-theory of the affine boundary quotient at a principal prime and located the class group “at the entrance”: for non-principal $\mathfrak{p}$ the affine monoid of Paper II (its §6) is not well defined over $\mathcal{O}_K$. This paper performs the repair and proves that the obstruction is itself the invariant. Let $h = \text{ord}([\mathfrak{p}] \in \text{Cl}_K)$ and $\mathfrak{p}^h = (\pi)$. The globally definable scalings form exactly the subsemigroup $h\mathbb{N}$ of the local valuation semigroup: a scaling of depth j exists over $\mathcal{O}_K$ if and only if $h \mid j$ (Theorem 1.2). The minimal repaired system is therefore the h -step system $C(\mathcal{O}_{\mathfrak{p}}) \rtimes (\mathcal{O}_K \rtimes_{\pi} \mathbb{N})$, well defined for every $\mathfrak{p}$, reducing to Paper XXX when $h = 1$; its boundary quotient has dilated form $C_0(K_{\mathfrak{p}}) \rtimes (\mathcal{O}_{K, \{\mathfrak{p}\}} \rtimes_{\pi} \mathbb{Z})$, and the entire exterior-tower Pimsner–Voiculescu computation of Paper XXX applies verbatim with (π, q^h) in place of (π, q) , $q = N\mathfrak{p}$. Consequences: the class of the unit generates a torsion summand of *exact* order $q^h - 1 = N\mathfrak{p}^h - 1$ — the order of $\mathfrak{p}$ in the class group survives in K_0 as the exact torsion order of $[1]$, and the published divisibility statement of Paper II must read $(N\mathfrak{p}^h - 1)[1] = 0$ in the repaired system; the intermediate layers contribute $\mathbb{Z}/|q^h \pm 1 - \text{Tr } \pi|'$ for quadratic fields, computed explicitly in $\mathbb{Q}(\sqrt{-5})$ at the non-principal prime above 3 ($K_0 \supseteq \mathbb{Z} \oplus \mathbb{Z}/8$ with $[1]$ of order 8, $K_1 \supseteq \mathbb{Z} \oplus \mathbb{Z}/2$). What the K-groups alone retain is the composite q^h ; separating q from h , the Galois descent from the Hilbert class field (where capitulation principalizes $\mathfrak{p}$ and $\text{Gal} \cong \text{Cl}_K$ acts), and the several-prime case are recorded as the remaining problems.

1. THE BOTTLENECK, AND THE SCALING-INDEX THEOREM

Fix a prime $\mathfrak{p}$ of K , $q = N\mathfrak{p}$, and let h be the order of $[\mathfrak{p}]$ in the class group Cl_K , so that $\mathfrak{p}^h = (\pi)$ is principal and no smaller positive power is. The affine monoid of Paper II (its §6) acts by $x \mapsto \gamma x + b$ with $b \in \mathcal{O}_K$ and γ a chosen uniformizing multiplier; the composition law requires $\gamma \mathcal{O}_K \subseteq \mathcal{O}_K$ with γ supported at $\mathfrak{p}$, i.e. $\gamma \in \mathcal{O}_K$ with $(\gamma) = \mathfrak{p}^j$ for some $j \geq 1$.

Lemma 1.1. *An element $\gamma \in \mathcal{O}_K$ with $(\gamma) = \mathfrak{p}^j$ exists if and only if $h \mid j$.*

Proof. $(\gamma) = \mathfrak{p}^j$ says $[\mathfrak{p}]^j$ is trivial in Cl_K , which is $h \mid j$; conversely $\pi^{j/h}$ works. □

Theorem 1.2 (The class group is the index of definable scalings). *The set of depths j for which the affine system of Paper II’s §6 admits a well-defined scaling of depth j over $\mathcal{O}_K$ is exactly $h\mathbb{N}$. The minimal well-defined affine system at $\mathfrak{p}$ is therefore the h -step system*

$$\mathcal{A}_{K, \{\mathfrak{p}\}}^{\text{aff}} := C(\mathcal{O}_{\mathfrak{p}}) \rtimes (\mathcal{O}_K \rtimes_{\pi} \mathbb{N}), \quad (\pi) = \mathfrak{p}^h,$$

with boundary quotient defined by the Cuntz relation over the q^h translates $\sum_{a \in \mathcal{O}_K/\mathfrak{p}^h} \mu_a \mu_a^ = 1$; it reduces to the system of Paper XXX when $h = 1$. Any choice of local uniformizer at intermediate depths $0 < j < h$ carries the translation module $\mathcal{O}_K$ to a fractional module in the ideal class $[\mathfrak{p}]^{-j} \neq 1$ and leaves the category of $\mathcal{O}_K$ -affine systems.*

Proof. Lemma 1.1 gives the depths. For the last statement, a local uniformizer $\varpi \in K_{\mathfrak{p}}$ of depth j maps the dense image of a fractional ideal $\mathfrak{a}$ to that of an ideal in the class $[\mathfrak{p}]^j[\mathfrak{a}]$ up to prime-to- $\mathfrak{p}$

Date: 5 September 2026.

Paper XXXI of the series on localized Bost–Connes systems. The framework is that of [1], cited by DOI; Papers II (published) and XXX of the series are referred to by numeral, and every fact used from them is restated in place.

global rescalings, which act by homeomorphisms of $K_{\mathfrak{p}}$; the marked class is an invariant of the pair (translation module, ambient $K_{\mathfrak{p}}$) modulo such rescalings, and it changes unless $h \mid j$. $\square$

Remark 1.3. Theorem 1.2 upgrades the observation of Paper XXX — “the class group is at the entrance” — to a structure theorem: the entrance fee is exactly h . Route B of the repair, passing to the Hilbert class field H where the extended ideal $\mathfrak{p}\mathcal{O}_H$ is principal by the principal ideal theorem [2, VI.7] and $\text{Gal}(H/K) \cong \text{Cl}_K$ acts, defines a one-step system upstairs whose Galois descent should remember the group Cl_K itself rather than only h ; its K-theory is Problem (ii) of §3.

2. K-THEORY OF THE REPAIRED BOUNDARY

Theorem 2.1. *Let $n = [K : \mathbb{Q}]$ and $N_{K/\mathbb{Q}}\pi = \pm q^h$. The dilated form of the repaired boundary is*

$$\partial\mathcal{A}_{K,\{\mathfrak{p}\}}^{\text{aff}} \otimes \mathbb{K} \cong C_0(K_{\mathfrak{p}}) \rtimes (\mathcal{O}_{K,\{\mathfrak{p}\}} \rtimes_{\pi} \mathbb{Z}),$$

where $\mathcal{O}_{K,\{\mathfrak{p}\}} = \bigcup_m \pi^{-m}\mathcal{O}_K$ is the ring of $\{\mathfrak{p}\}$ -integers, dense in $K_{\mathfrak{p}}$; and the exterior-tower computation of Paper XXX applies verbatim with q^h in place of q : with $M_r = \varinjlim (\Lambda^r \mathcal{O}_K, t_r)$, $t_r = \pm(\Lambda^{n-r}\pi)^{\vee}$, $\alpha_* = q^{-h}\Lambda^r\pi$, and $T_r = \text{coker}(1 - \alpha_*|M_r)$, one has, when $N\pi = +q^h$ and $\pi^A \neq q^h$ for all proper archimedean subsets A ,

$$K_0 \cong \mathbb{Z} \oplus \bigoplus_{r \text{ even}} T_r, \quad K_1 \cong \mathbb{Z} \oplus \bigoplus_{r \text{ odd}} T_r, \quad T_0 = \mathbb{Z}/(q^h - 1) \text{ generated by } [1],$$

(the two $\mathbb{Z}$'s replaced by a single $\mathbb{Z}/2$ in parity n when $N\pi = -q^h$), the orders of the middle layers being the regularized $|\det(q^h \cdot 1 - \Lambda^r \pi)|$ as in Paper XXX. In particular:

- (1) The order of $\mathfrak{p}$ in the class group survives in K_0 : the class $[1]$ has exact order $N\mathfrak{p}^h - 1$. The divisibility statement of Paper II's §6 must accordingly read $(N\mathfrak{p}^h - 1)[1] = 0$ for non-principal $\mathfrak{p}$, and it is attained.
- (2) For K quadratic, $K_1(\partial\mathcal{A}^{\text{aff}}) \supseteq \mathbb{Z}/|q^h \pm 1 - \text{Tr } \pi|'$ (prime-to- q part), the sign that of $N\pi$.

Proof. The dilation, the tower, the two calibrations and the Pimsner–Voiculescu [3] assembly are those of Paper XXX, restated there in full; the only inputs were a global generator of the scaling ideal and its norm, here π and $N(\mathfrak{p}^h) = q^h$. The exactness of the order of $[1]$ is the $r = 0$ layer: $1 - \alpha_* = 1 - q^{-h}$ on $\mathbb{Z}[1/q]$ has cokernel $\mathbb{Z}/(q^h - 1)$ generated by the class of $1_{\mathcal{O}_{\mathfrak{p}}}$. $\square$

Example 2.2 ($\mathbb{Q}(\sqrt{-5})$ at the non-principal prime above 3). $K = \mathbb{Q}(\sqrt{-5})$, $\text{Cl}_K \cong \mathbb{Z}/2$, $\mathfrak{p}_3 = (3, 1 + \sqrt{-5})$ non-principal, $h = 2$, $q = 3$, and $\mathfrak{p}_3^2 = (\pi)$ with $\pi = 2 - \sqrt{-5}$, $N\pi = 9 = q^h$, $\text{Tr } \pi = 4$. Then: $[1]$ has exact order $q^h - 1 = 8$, so $K_0 \supseteq \mathbb{Z} \oplus \mathbb{Z}/8$ with $[1]$ generating the torsion; and the middle layer gives $N(q^h - \pi) = N(7 + \sqrt{-5}) = 54 = 2 \cdot 27$, of prime-to-3 part 2, so $K_1 \supseteq \mathbb{Z} \oplus \mathbb{Z}/2$, consistently with $|q^h + 1 - \text{Tr } \pi| = |9 + 1 - 4| = 6$ of prime-to-3 part 2. Had $\mathfrak{p}_3$ been principal the order of $[1]$ would have been $q - 1 = 2$: the K-theory separates the two situations.

Remark 2.3 (Separating q from h : the middle layers generically do). The order of $[1]$ sees only the composite q^h , and pairs realizing the same composite exist: $\mathbb{Q}(\sqrt{-5})$ at $\mathfrak{p}_3$ ($q = 3$, $h = 2$) against $\mathbb{Q}(i)$ at the inert prime (3) ($q' = 9$, $h' = 1$, principal) both have $\text{ord}[1] = 8$. But the middle layers separate them: for $\mathbb{Q}(i)$, $\pi = 3$ acts as the scalar $\frac{1}{3}$ on $M_1 \cong \mathbb{Z}[1/3]^2$, so $T_1 = \text{coker}(\frac{2}{3})^{\oplus 2} \cong (\mathbb{Z}/2)^2$ of order 4, against the $\mathbb{Z}/2$ of Example 2.2. So Problem (i) of §3 is partially resolved: the K-groups themselves generically separate (q, h) through the trace data of the intermediate exterior layers, and only pairs with matching characteristic data at every layer remain for the dual coaction to distinguish.

Remark 2.4 (Unit-orbit sensitivity persists). As in Paper XXX, replacing π by $u\pi$ for a unit u of norm 1 changes the middle layers; for imaginary quadratic fields the unit group is finite and

the ambiguity is a root of unity, so the repaired invariant is canonical there up to finitely many choices — a small unexpected bonus of the imaginary quadratic case.

3. THE ARITHMETIC VERDICT, AND WHAT REMAINS

The class group enters the K-theory as *torsion, not rank*: the free ranks of Theorem 2.1 are those of Paper XXX, unchanged, while the order of the distinguished class [1] is promoted from $q - 1$ to $q^h - 1$. Three problems remain. (i) *Separating q from h* : partially resolved by Remark 2.3 — the middle layers generically separate; what remains is the exceptional locus of pairs with matching layer data, where the dual coaction is the candidate. (ii) *Galois descent from the Hilbert class field*: capitulation principalizes $\mathfrak{p}$ upstairs with $\text{Gal}(H/K) \cong \text{Cl}_K$; the equivariant K-theory of the descended system should remember Cl_K itself, and its computation — a genuine descent spectral sequence, not an iterated Pimsner–Voiculescu — is open. (iii) *Several primes*: for $|S| \geq 2$ the definable scalings form the kernel of $\mathbb{Z}^S \rightarrow \text{Cl}_K$, $e \mapsto \prod \mathfrak{p}^{e_{\mathfrak{p}}}$, a finite-index subgroup whose index divides $|\text{Cl}_K|$ and whose K-theoretic footprint, layer by layer, is the natural continuation of this paper.

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